

How a channel's changing shape slows diffusion

The transverse cell problem, explained through examples

Imagine three long channels with exactly the same area at every axial position. In one, every cross section is the same disc. In another, an ellipse turns as we move along the channel. A third changes from a disc into a three-lobed shape while preserving its area. Would a diffusing particle spread along all three at the same rate?

The area alone suggests no difference. The reflecting walls suggest otherwise: a sloping wall couples longitudinal and transverse motion. This tension is the starting point of the paper.

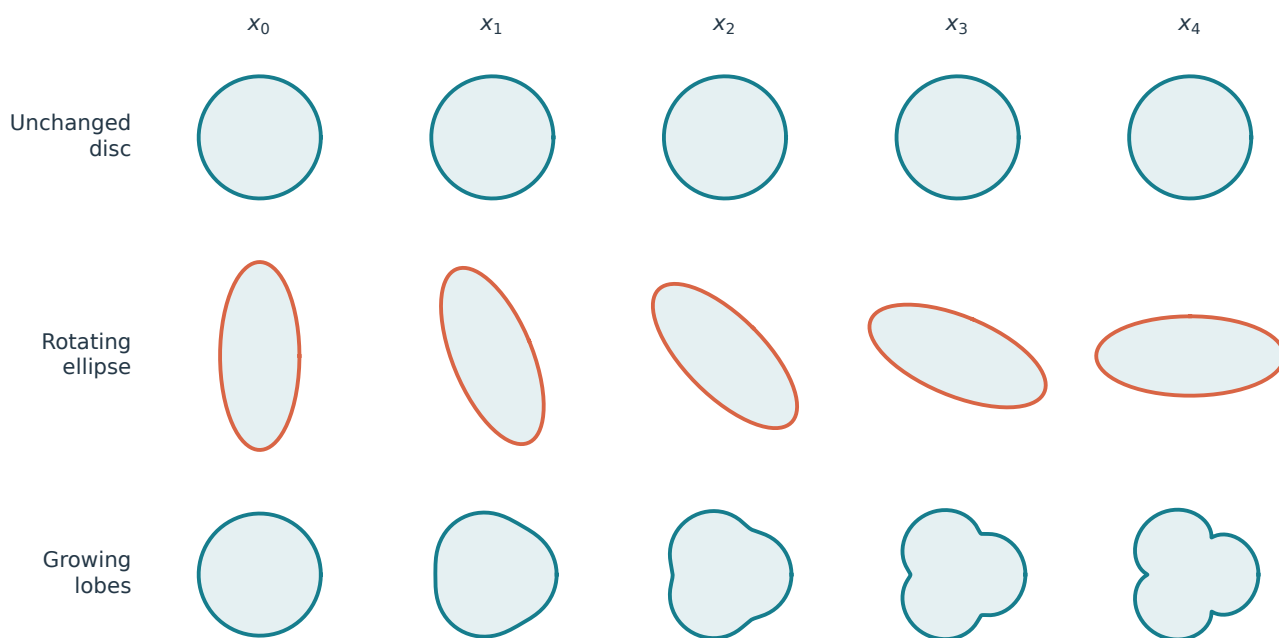


Figure 1. Three sequences of sections. Every outlined region has area π . The horizontal ordering is spatial: these are different positions along a fixed channel, not snapshots of a moving wall.

The point to remember. The first correction is determined by an energy of *how the section changes*, rather than by its area alone.

These notes are for mathematicians who know multivariable calculus, the divergence theorem and the Laplacian, but have not worked on diffusion in channels. They can be read as a short paper or used as the narrative for a 40–50 minute talk. We first build the local picture, then examine translation, twist and lobes, and finally explain a rigorous result for periodic channels.

This is a separate companion to the corrected manuscript *Effective diffusion in channels of varying cross section at finite transversal rate: the transverse cell problem and its gradient formulation* by Carlos Valero Valdés. Full statements and proofs remain there. The distinction between a conditional approximation and a proved periodic bound will remain visible throughout.

1 What are we trying to reduce?

In the main paper: Sections 1–3; Eqs. (1), (3), (6)–(11).

A channel is a family of planar regions indexed by one spatial coordinate:

$$\Theta = \{(x, y) : y \in \Omega(x)\}, \quad y = (y_1, y_2), \quad A(x) = |\Omega(x)|.$$

The region is fixed in time. A particle diffuses inside it and reflects at the wall. Let $P(x, y, t)$ be its density per unit volume, and let

$$p(x, t) = \int_{\Omega(x)} P(x, y, t) dy, \quad f(x, t) = \frac{p(x, t)}{A(x)}.$$

Here p is density per unit axial length; f is the section-averaged volume density. If equilibrium makes P constant, then p is proportional to A : wider slices hold more particles even when the volume density is uniform.

We distinguish longitudinal and transverse diffusivities:

$$D_{\parallel} = D_0, \quad D_{\perp} = \frac{D_0}{\varepsilon}, \quad P_t = D_0 P_{xx} + \frac{D_0}{\varepsilon} \Delta_y P.$$

Small ε means fast transverse mixing at fixed geometry. For physical length scales ℓ across the channel and L along it, the ratio of mixing times is of order $\varepsilon \ell^2 / L^2$; section shape also matters.

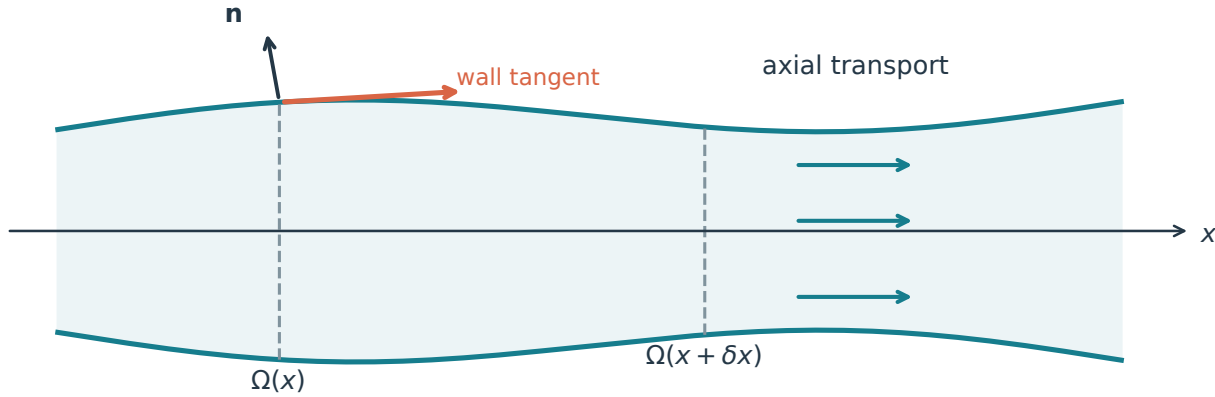


Figure 2. A planar schematic of the fixed wall. The outward three-dimensional normal has an axial component where the wall slopes. Reflection therefore couples axial and transverse derivatives.

If transverse mixing were instantaneous, P would be almost constant on each section. The resulting one-dimensional equation is

$$p_t = \partial_x [D_0 A \partial_x (p/A)].$$

This is the leading Fick–Jacobs equation. Our question is how its coefficient changes when transverse equilibration takes a finite time.

The point to remember. The quantity that drives flux is the gradient of $f = p/A$. A gradient of p alone would incorrectly predict flow at equilibrium in a channel of varying area.

2 A Neumann problem hidden in the wall

In the main paper: Proposition 1 and Eqs. (4), (5), (7), (12).

Describe the boundary of a section by $\gamma(x, s)$ and let ν be its outward planar unit normal. The scalar

$$v_n = \partial_x \gamma \cdot \nu$$

measures how the boundary changes *as we move along x* . Despite the notation, it is not a velocity in time. Tangential changes merely relabel the boundary; v_n records the geometrically relevant change.

The full wall normal is proportional to $(-v_n, \nu)$. Requiring the diffusive flux to have zero normal component gives

$$\partial_\nu P = \varepsilon v_n P_x.$$

A density depending only on x generally fails this condition. A small transverse correction must supply the missing normal derivative.

At each station, the paper introduces the mean-zero function χ :

$$\Delta_y \chi = \frac{A'}{A}, \quad \partial_\nu \chi = v_n, \quad \int_{\Omega(x)} \chi \, dy = 0.$$

This is the *transverse cell problem*: one elliptic boundary-value problem on a single section. The name does not require periodicity.

Why is it solvable? Integrating the equation and using the divergence theorem asks for

$$\int_{\Omega} \frac{A'}{A} \, dy = A' = \oint_{\partial\Omega} v_n \, ds.$$

The last equality is exactly the formula for the derivative of the area. The geometric compatibility condition is built in. On a smooth bounded connected section, standard Neumann theory then gives χ , unique after fixing its mean.

The right side can vanish without the boundary datum vanishing. If a section changes shape at constant area, positive and negative values of v_n cancel around the boundary. They do not cancel pointwise, so χ can remain nonzero.

The point to remember. Constant area means zero *net* normal change. It does not mean that every part of the boundary stays in place. This is why area-preserving twist and lobe formation can affect diffusion.

3 The energy that measures shape change

In the main paper: Proposition 1, Section 5; Eqs. (13), (20).

Set $u = \nabla_y \chi$ and define the normalized Dirichlet energy

$$E(x) = \frac{1}{A(x)} \int_{\Omega(x)} |u|^2 dy.$$

The field u measures the leading transverse adjustment relative to the axial flux density. It is not an imposed fluid velocity: the model has diffusion and reflection, with no advecting flow.

An integration by parts makes the energy accessible from the wall:

$$\int_{\Omega} |\nabla_y \chi|^2 = \oint_{\partial\Omega} \chi v_n ds - \frac{A'}{A} \int_{\Omega} \chi dy = \oint_{\partial\Omega} \chi v_n ds.$$

Thus AE is a boundary pairing. Differentiating the normalization of χ gives another useful identity:

$$0 = \frac{d}{dx} \int_{\Omega(x)} \chi = \int_{\Omega} \chi_x + \oint_{\partial\Omega} \chi v_n, \quad \boxed{\int_{\Omega} \chi_x = -AE.}$$

This minus sign is where the first-order reduction in longitudinal transport enters.

There is also a useful energy-minimizing interpretation. Among smooth fields q with

$$\operatorname{div} q = A'/A, \quad q \cdot \nu = v_n,$$

the gradient field u has smallest squared L^2 norm. Indeed, $q = u + v$ with $\operatorname{div} v = 0$ and $v \cdot \nu = 0$, so

$$\int_{\Omega} u \cdot v = 0, \quad \int_{\Omega} |q|^2 = \int_{\Omega} |u|^2 + \int_{\Omega} |v|^2.$$

The cell energy measures the least transverse adjustment compatible with the prescribed normal change.

On a simply connected section, we can often find u directly:

$$\operatorname{div} u = A'/A, \quad \operatorname{curl} u = 0, \quad u \cdot \nu = v_n.$$

On a section with holes, zero circulation around each hole must also be imposed. The scalar formulation already enforces it.

The point to remember. $E \geq 0$, and $E = 0$ exactly when $v_n = 0$ everywhere. A mere relabelling costs nothing. Actual normal change has positive energy, however well its positive and negative parts balance.

4 From the cell energy to a transport law

In the main paper: Theorem 1, Remark 1 and Section 4.1; Eqs. (14)–(19).

Here is the local result in its useful form:

$$D^{[1]}(x) = D_0[1 - \varepsilon E(x)].$$

The superscript [1] means *first-order approximation*. It is not a formula claimed to be exact at finite ε . The hypothesis matters. Suppose the actual density has a controlled expansion

$$P_\varepsilon = P_0 + \varepsilon P_1 + O(\varepsilon^2)$$

up to the wall, with uniform control of the spatial and time derivatives needed in the PDE. At order zero, P_0 is harmonic in the section with zero normal derivative, hence $P_0 = f_0(x, t)$. At the next order, the Neumann compatibility condition gives the leading Fick–Jacobs equation, and the cell problem yields

$$P_1 = a_1(x, t) + (f_0)_x \chi(x, y).$$

The function a_1 is an axial mean correction. Fixing the mean of χ does not give permission to set a_1 to zero.

The rest of the derivation is short. Write $Q_\varepsilon = \int_{\Omega(x)} (P_\varepsilon)_x dy$. Then

$$\begin{aligned} Q_\varepsilon &= A(f_0)_x + \varepsilon \left[A(a_1)_x + (f_0)_x \int_{\Omega} \chi_x \right] + O(\varepsilon^2) \\ &= A(f_0)_x + \varepsilon [A(a_1)_x - AE(f_0)_x] + O(\varepsilon^2) \\ &= A(1 - \varepsilon E)(f_\varepsilon)_x + O(\varepsilon^2), \end{aligned}$$

because the actual mean is $f_\varepsilon = f_0 + \varepsilon a_1 + O(\varepsilon^2)$. The integrated physical flux is $-D_0 Q_\varepsilon$, and exact conservation says $(p_\varepsilon)_t = D_0(Q_\varepsilon)_x$. These identities give the reduced equation with coefficient $D^{[1]}$ and an $O(\varepsilon^2)$ remainder.

What this proves. It identifies the coefficient when the regular expansion exists. The full hypothesis in the paper controls two spatial derivatives and one time derivative of the remainder. It does not establish that arbitrary initial and end data produce such an expansion.

The point to remember. The local calculation is conditional but geometrically explicit. Later, periodicity will let us prove a first-order *long-time* result without assuming this density expansion.

An initially nonuniform transverse profile needs time to relax. Incompatible end data can create a thin axial layer. Neither effect is removed merely by drawing a smooth channel.

5 Two examples you can solve by inspection

In the main paper: Section 5 and Example 1; Eq. (27).

Translation. Let $\Omega(x) = \Omega_0 + c(x)$, with a fixed shape and a changing position. Its area is constant and $v_n = c' \cdot \nu$. The constant field

$$u = c'(x)$$

has zero divergence and curl, and the correct normal component. Consequently

$$E = |c'|^2.$$

Even a circular section produces a correction if its centre wanders relative to the chosen straight axial coordinate.

Dilation. Let $\Omega(x) = R(x)\Omega_0$. The field

$$u = \frac{R'}{R} y$$

has divergence $2R'/R = A'/A$ and the correct boundary trace. Thus

$$E = \left(\frac{R'}{R}\right)^2 \langle |y|^2 \rangle.$$

For a disc of radius R , $\langle |y|^2 \rangle = R^2/2$, giving $E = R'^2/2$. Its first-order coefficient is $D_0(1 - \varepsilon R'^2/2)$.



Figure 3. Cell-gradient fields on representative discs. Translation requires a constant transverse adjustment; dilation requires a radial one. Arrows depict u , not particle trajectories or a fluid flow.

These formulas have a common feature: the rate of geometric change enters quadratically. Reverse the direction of translation or dilation and the energy is unchanged. Multiply its rate by two and E increases by four.

When a shape translates, dilates about its centroid and rotates, linearity adds the corresponding cell fields. In the setting of Example 3 of the main paper, the three fields are mutually orthogonal in L^2 , so their energies add as well. This is a geometric Pythagorean theorem behind the more elaborate helical examples.

The point to remember. A useful way to compute χ is to avoid computing χ : guess its gradient, verify divergence, curl and boundary trace, then integrate its squared norm.

6 Why a rotating ellipse matters

In the main paper: Example 2 and Section 10.1; Eq. (28).

A disc rotated about its centre is the same disc. Its boundary points move tangentially, so $v_n = 0$ and $E = 0$. An ellipse is different: turning it displaces parts of its boundary in the normal direction.

Let the semiaxes be a, b and let the orientation angle be $\theta(x)$. In coordinates aligned with the ellipse at a given station, put

$$\kappa = \theta' \frac{a^2 - b^2}{a^2 + b^2}, \quad \chi = \kappa y_1 y_2, \quad u = \kappa(y_2, y_1).$$

The potential is harmonic. Matching its normal derivative to the rotational wall datum fixes κ . Since $\langle y_1^2 + y_2^2 \rangle = (a^2 + b^2)/4$,

$$E = \frac{\theta'^2 (a^2 - b^2)^2}{4(a^2 + b^2)}.$$

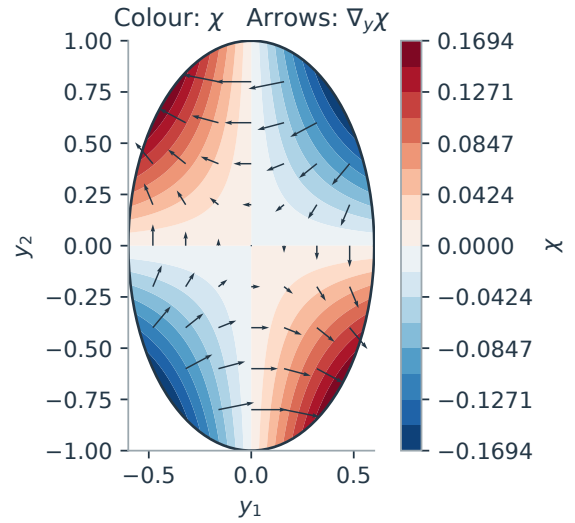
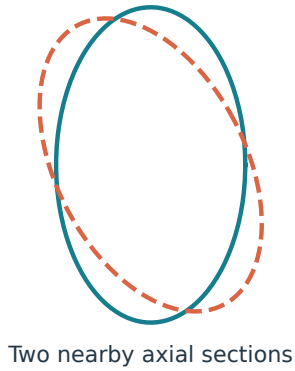


Figure 4. Left: two orientations of the same ellipse. Right: its harmonic corrector and gradient for $a = 0.6$, $b = 1$, $\theta' = 1.2$. The sign-changing corrector has zero mean, but its squared-gradient energy is positive.

Explore in 3D: Twisting ellipse.

The formula passes three immediate tests. There is no correction when the ellipse stops turning; its sign is independent of clockwise or counterclockwise twist; and the correction vanishes as $a = b$. No area gradient is involved.

For uniform twist, the channel is periodic after a half-turn. The periodic theorem will show that this same E is the first-order loss of the long-time diffusivity. For $a = 0.6$, $b = 1$ and $\theta' = 1.2$, the value is approximately 0.108424.

The point to remember. The observable feature is not a painted mark rotating on the wall. It is the rotation of a noncircular *region*. The formula correctly ignores the rotation of a disc.

7 Shape change without translation or rotation

In the main paper: Example 4 and Section 10.2; Eqs. (40), (41), (48).

Consider the image of the unit disc under the polynomial map

$$\varphi(\zeta) = R(\zeta + m\zeta^4), \quad R = \frac{1}{\sqrt{1+4m^2}}, \quad 0 \leq m < \frac{1}{4}.$$

The area is always π . As m increases, three lobes emerge. Threefold symmetry fixes the centroid at the origin; this is a deformation of the shape itself.

The paper's conformal formula gives

$$E(x) = \frac{\dot{m}^2(1-16m^2)^2}{3(1+4m^2)^4}, \quad \dot{m} = \frac{dm}{dx}.$$

It depends on both the shape m and the rate $\dot{m}$ at which that shape changes along the axis.

For a periodic example, take $m(x) = 0.22 \sin^2(\pi x/L)$ and $L = 1$. The tube expands into lobes and then returns to a disc:

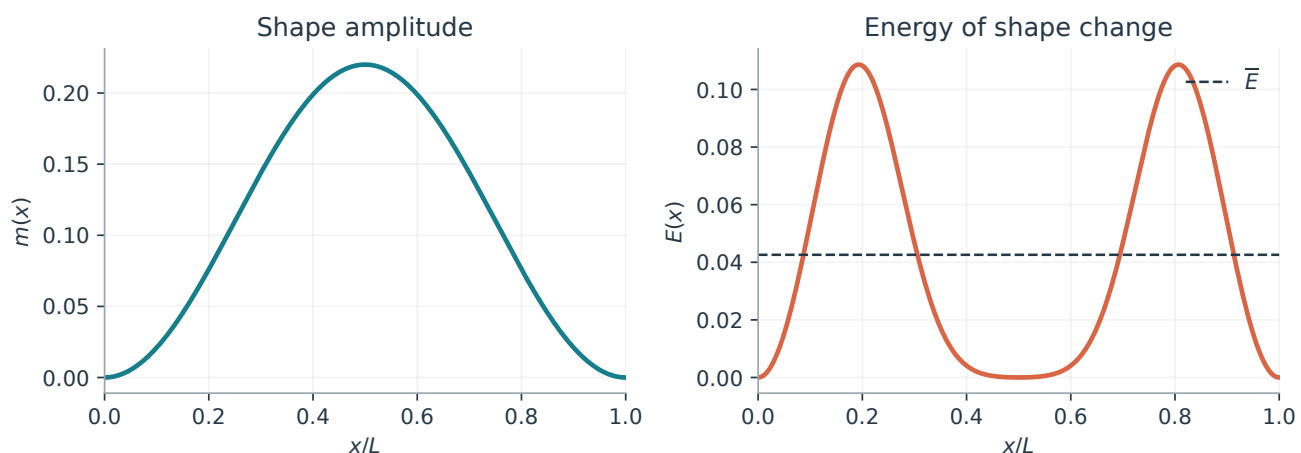


Figure 5. The largest deformation occurs at $x/L = 1/2$, precisely where the shape stops changing to first order and $E = 0$. The period average is $\bar{E} \approx 0.0425983 > 0$. **Explore in 3D: Constant-area lobes.**

This distinguishes two questions that are easy to confuse. How far is a section from a disc? How much does it change as x increases? The energy answers the second question, with a coefficient that depends on the present shape. A fixed, highly noncircular section would give a straight cylinder and $E = 0$ everywhere.

There is a striking limit. As $m \uparrow 1/4$ with bounded $\dot{m}$, the formula for E tends to zero even though the boundary develops cusps. The geometry explains the small energy: the normal-flux measure tends to zero. This is a limiting computation, not permission to apply a smooth-boundary theorem at a cusp.

The point to remember. The local energy can vanish at some stations while its period average is positive. A local coefficient and a long-time coefficient therefore answer different questions even at constant area.

8 Why complex analysis makes the example finite

In the main paper: Sections 6 and 8; Proposition 3, Eqs. (22)–(24), (33)–(39).

Identify a section with a domain in the complex plane and write

$$z = y_1 + iy_2, \quad w = u_1 - iu_2.$$

The divergence and curl equations combine into

$$\partial_{\bar{z}} w = \frac{A'}{2A}.$$

Thus $w = \alpha \bar{z} + h(z)$, where $\alpha = A'/(2A)$ and h is holomorphic. At constant area, w itself is holomorphic.

Now map the unit disc onto the section by φ and define $G = (h \circ \varphi)\varphi'$. At constant area, conformal change of variables cancels the Jacobian against the derivative factor:

$$\int_{\Omega} |w|^2 dy = \int_{\mathbb{D}} |G|^2 dA_{\zeta}.$$

If $G(\zeta) = \sum_{k \geq 0} g_k \zeta^k$, orthogonality of monomials on the disc gives

$$\int_{\mathbb{D}} |G|^2 dA_{\zeta} = \pi \sum_{k \geq 0} \frac{|g_k|^2}{k+1}.$$

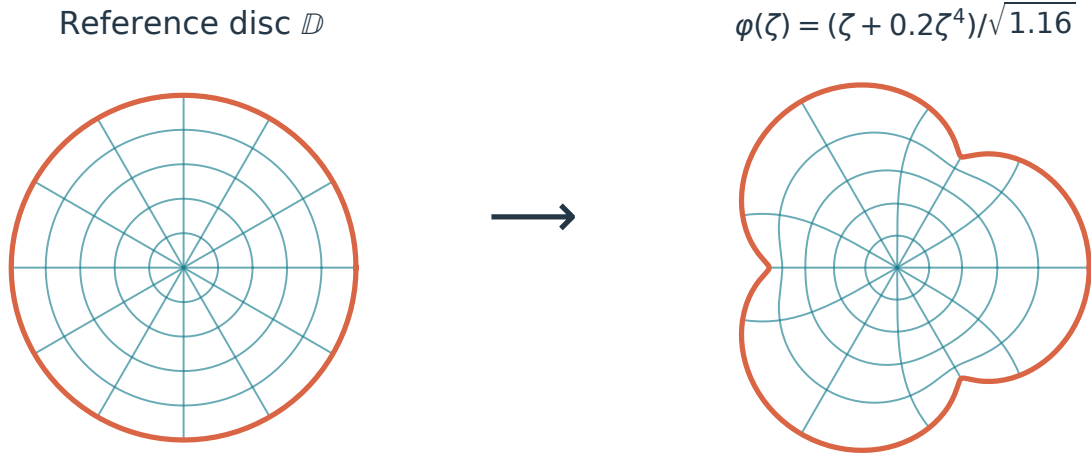


Figure 6. A polar grid carried from the disc into a three-lobed section by an injective analytic map.

Explore in 3D: Conformal map.

For a polynomial map, the transformed boundary datum has finitely many Fourier modes. Matching their real parts makes $H = \zeta G$ a polynomial as well. More explicitly, if $\varphi = \sum_{k=1}^M c_k \zeta^k$, set

$$\hat{c}_k = \dot{c}_k - \frac{A'}{2A} c_k, \quad K_n = \sum_k k c_k \overline{\hat{c}_{k-n}}, \quad h_n = K_n + \overline{K_{-n}}.$$

Then $H = \sum_{n \geq 1} h_n \zeta^n$ and, at constant area, $E = (\pi/A) \sum_{n \geq 1} |h_n|^2/n$. Coefficients outside the range are zero.

The point to remember. Finiteness lives on the reference disc. The physical-plane corrector need not be polynomial, and one never has to invert φ explicitly to obtain the energy.

9 Three diffusion coefficients, three experiments

In the main paper: Section 9.1 and Section 11; Eqs. (47), (49).

Before comparing formulas, we must say what is being measured. The main paper keeps three quantities separate.

Quantity	Question	Mathematical object
$D^{[1]}(x)$	What local flux law approximates a transversely relaxed density?	$D_0(1 - \varepsilon E(x))$, under the regular-expansion hypothesis
$D_{\text{st}}[P](x)$	What flux-to-mean-gradient ratio does one specified steady solution have?	$D_0 Q/[A(p/A)']$, where the denominator is nonzero
D_{long}	At what rate does a particle cloud spread over many repeated cells?	A constant periodic homogenized diffusivity

For the periodic channel, the last quantity can be interpreted as

$$D_{\text{long}} = \lim_{t \rightarrow \infty} \frac{\text{Var}(X_t)}{2t},$$

where X_t is the unwrapped axial position of the reflected diffusion. It averages the cumulative effect of many sections. It is not a value attached to one station.

A familiar one-dimensional calculation already illustrates the difference. Suppose a reduced model has a positive periodic local coefficient $d(x)$. Its large-scale coefficient is

$$D_{\text{long}}^{\text{1D}} = \frac{L^2}{\left(\int_0^L A(x) dx \right) \left(\int_0^L \frac{dx}{d(x)A(x)} \right)}.$$

The second integral is a resistance: for a steady flux, the gradient is inversely proportional to dA . At constant area, the answer is the harmonic mean of d , not its value at a selected x .

Even when $d = D_0$ is constant, variable area gives

$$\frac{D_{\text{long}}^{\text{1D}}}{D_0} = \frac{1}{\overline{A A^{-1}}} \leq 1.$$

The inequality is Cauchy–Schwarz. This is the large-scale effect of the area profile before adding finite transverse relaxation.

The point to remember. An exact answer to one of these three questions is not automatically an exact answer to the other two. Keeping the observable fixed is part of the mathematics.

10 A rigorous periodic theorem

In the main paper: Theorem 2; Eqs. (42)–(45).

Assume the section family is smooth and periodic with period L . Write $V = \int_0^L A(x) dx$, use an overbar for the axial average $L^{-1} \int_0^L$, and set

$$\langle F \rangle_V = \frac{1}{V} \int_{\Theta_{\text{per}}} F dV.$$

Periodic homogenization expresses the normalized diffusivity as

$$d_\varepsilon := \frac{D_{\text{long}}}{D_0} = \min_{w \text{ periodic}} \langle (1 + w_x)^2 + \varepsilon^{-1} |\nabla_y w|^2 \rangle_V.$$

The function $x + w$ is a corrected axial coordinate: it gains L over one period while adjusting to the reflecting walls. The minimization is a Dirichlet principle.

All the geometry needed for the following bounds comes from χ :

$$\mathcal{A} = \overline{A}, \quad B = \overline{A^{-1}}, \quad C = \overline{E/A}, \quad g = \frac{1}{BA}, \quad S = \langle |\partial_x(g\chi)|^2 \rangle_V.$$

The theorem proves, for every $\varepsilon > 0$,

$$\frac{1}{\mathcal{A}(B + \varepsilon C)} \leq d_\varepsilon \leq \frac{1}{\mathcal{A}B} - \varepsilon \frac{C}{\mathcal{A}B^2} + \varepsilon^2 S.$$

In particular, defining the first-order expression $d_{\text{lin}} = 1/(\mathcal{A}B) - \varepsilon C/(\mathcal{A}B^2)$, we obtain

$$0 \leq d_\varepsilon - d_{\text{lin}} \leq \varepsilon^2 S.$$

This is an explicit remainder bound at fixed geometry. No regular expansion of a density has been assumed.

The coefficient has two understandable pieces:

$$d_\varepsilon = \underbrace{\frac{1}{\overline{A A^{-1}}}}_{\text{leading area effect}} \left[1 - \varepsilon \underbrace{\frac{\overline{E/A}}{\overline{A^{-1}}}}_{\text{resistance-weighted cell energy}} \right] + O(\varepsilon^2).$$

Narrow sections receive greater weight through $1/A$. At constant area, this simplifies to $1 - \varepsilon \overline{E} + O(\varepsilon^2)$.

The point to remember. The local theorem needs a hypothesis about the solution. The periodic theorem obtains its expansion by placing the exact coefficient between two explicit competitors.

11 The proof uses one potential and one flux

In the main paper: Proof of Theorem 2 and Corollary 1.

A variational minimum gives an upper bound whenever we insert an admissible function. Choose $f'_0 = g = 1/(BA)$; since $\bar{g} = 1$, $f_0 - x$ is periodic. The trial function

$$w_{\text{trial}} = f_0 - x + \varepsilon g \chi$$

has the leading axial profile and its transverse correction. Expanding its energy and using $\int_{\Omega} \chi_x = -AE$ gives exactly $d_{\text{lin}} + \varepsilon^2 S$.

For a lower bound, use the vector field

$$j = \frac{1}{A}(1, \nabla_y \chi).$$

Its divergence is zero because $\partial_x(A^{-1}) + A^{-1}\Delta_y \chi = 0$. It is tangent to the wall because its normal component is proportional to $-v_n + \partial_\nu \chi = 0$. Thus integration by parts removes its pairing with any periodic gradient, and weighted Cauchy–Schwarz gives the reciprocal lower bound. The same transverse corrector supplies both competitors.

At constant area, optimizing the trial family $w = \alpha \chi$ improves the upper bound to

$$\frac{1}{1 + \varepsilon \bar{E}} \leq d_\varepsilon \leq 1 - \frac{\varepsilon \bar{E}^2}{\bar{E} + \varepsilon T}, \quad T = \langle \chi_x^2 \rangle_V,$$

when $\bar{E} > 0$. These bounds are plotted below.

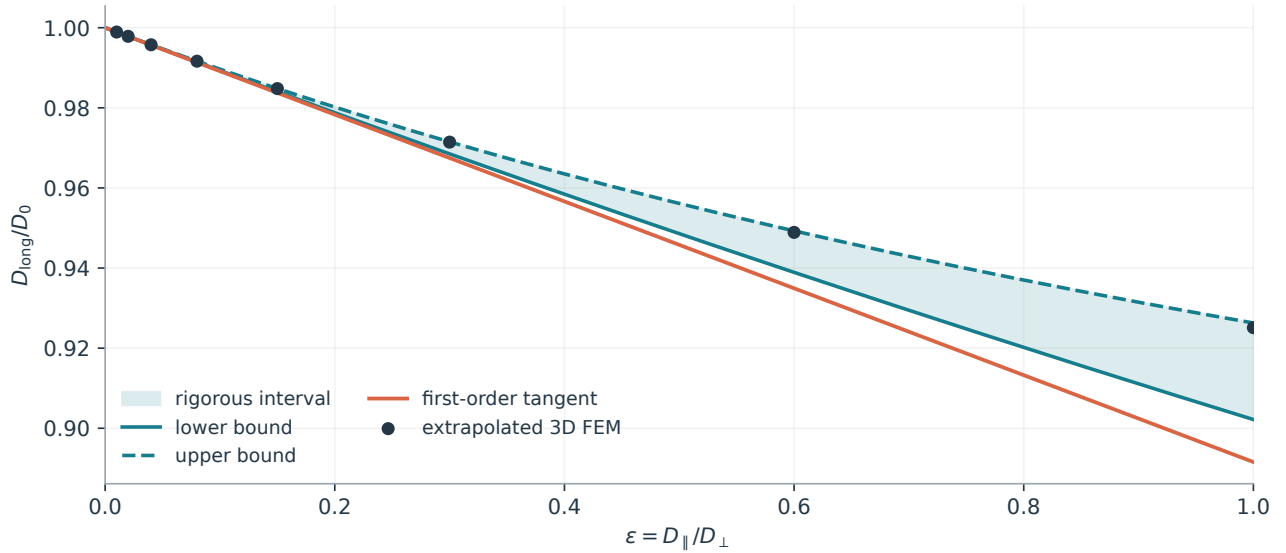


Figure 7. Uniformly twisting ellipse, $a = 0.6$, $b = 1$, $\theta' = 1.2$. The shaded region lies between rigorous continuum bounds. Dots are the main paper’s cached, extrapolated three-dimensional FEM values, not certified error bars.

Explore in 3D: Periodic bounds.

Here E is constant. With $\kappa = \theta'(a^2 - b^2)/(a^2 + b^2)$, differentiation at fixed physical y gives $T = \kappa^2 \theta'^2 [(a^4 + b^4)/8 - a^2 b^2/12]$. The exact coefficient lies above its first-order tangent. A first-order formula describes the slope at $\varepsilon = 0$, not the entire curve.

12 An exact solution is an instructive test

In the main paper: Example 5, Proposition 4 and Section 11.3.

For a cone $\Omega(x) = x\Omega_1$, $x > 0$, the function

$$P = (x^2 + \varepsilon|y|^2)^{-1/2}$$

solves both the anisotropic steady equation and the reflecting wall condition. It is a reciprocal-distance potential in rescaled coordinates. For a circular cone of slope λ ,

$$\frac{D_{\text{st}}[P]}{D_0} = \frac{1}{\sqrt{1 + \varepsilon\lambda^2}} = 1 - \frac{\varepsilon\lambda^2}{2} + \frac{3\varepsilon^2\lambda^4}{8} + \dots$$

The first term agrees with the dilation energy $E = \lambda^2/2$.

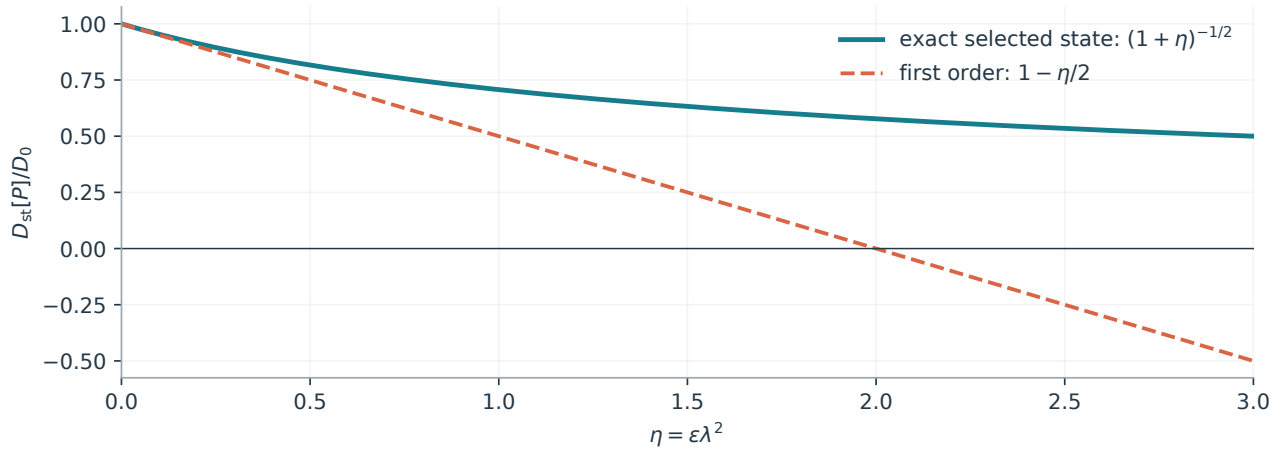


Figure 8. An exact selected steady-state ratio and its tangent. A negative first-order truncation at large η signals the failure of that approximation, not negative physical diffusion.

This example also catches a subtle normalization mistake. For the unit-slope cone,

$$P_1 = -\frac{|y|^2}{2x^3}, \quad \chi = \frac{|y|^2 - x^2/2}{2x}, \quad f_0 = \frac{1}{x}.$$

The equality is $P_1 = -1/(4x) + f'_0\chi$. The mean correction is real information about the solution; it cannot be removed by normalizing χ .

A second exact example is the oblique cylinder $\Omega(x) = \Omega_0 + xv$. Its affine solution $P = x + \varepsilon v \cdot y$ gives $D_{\text{st}}[P]/D_0 = (1 + \varepsilon|v|^2)^{-1}$, agreeing at first order with $E = |v|^2$.

There is even a specified local steady state for which $D_{\text{st}}[P]/D_0 = 100/97 > 1$ (Main §11.3). This does not contradict $E \geq 0$ or the periodic bound $D_{\text{long}} \leq D_0$. A pointwise ratio in a selected state is a different observable.

The point to remember. Exact solutions test a derivation exceptionally well. They still need their state and boundary conditions attached: an exact steady ratio is not an exact transient reduction.

13 What the picture explains, and where it stops

In the main paper: Remarks 1–3, Section 10 and the conclusion.

We can now return to the opening question. Equal area fixes the leading equilibrium weight, but it leaves many ways for a section to change. The normal boundary motion determines a Neumann problem, and the problem’s energy measures the first loss of axial transport.

This is the paper’s unifying step. Translation, dilation, twist and non-affine deformations fit the same cell problem. Polynomial conformal maps make an important class computable by finite algebra. Periodic variational bounds turn that cell information into a rigorous large-scale expansion.

The field already has systematic Fick–Jacobs corrections, variational approaches, planar curved-midline reductions and symmetric three-dimensional examples. The contributions here concern general smooth section deformations, the explicit conformal energy calculation, and the pair of periodic bounds. The introduction to the main paper compares these with the relevant literature.

Three limits of the argument deserve attention.

First, transverse equilibration can be slow in a section with a narrow connecting neck. The first positive Neumann eigenvalue μ_1 controls the slowest transverse mode. The relevant comparison involves $\varepsilon/(\mu_1 L^2)$, not diameter alone. Likewise, nonuniform end data can produce a layer of length $\sqrt{\varepsilon/\mu_1}$.

Second, an energy formula at first order does not by itself bound every higher-order correction. Geometric derivatives enter later terms. A small value of εE is useful information, but is not a universal accuracy guarantee over arbitrary families of channels.

Third, numerical and symbolic checks answer narrower questions than theorems. The main paper compares three-dimensional FEM computations with the analytic energies, performs refinements, and uses an independent two-dimensional reduction for uniform twist. The separate SymPy annex checks 45 exact residual identities. Neither a small solver residual nor an exact algebraic cancellation proves a missing analytic hypothesis.

The point to remember. The useful division of labour is clear: geometry supplies the corrector; analysis states the regime and proves bounds; exact examples and computations test the formulas.

Natural next mathematical problems are to establish the local expansion from specified initial and end data, obtain estimates uniform over controlled shape families, and optimize shape changes subject to geometric constraints. The explicit periodic bounds provide concrete quantities with which to begin.

14 A route through the talk and the paper

Suggested 45-minute route. Start with Figure 1 and let the equal-area question motivate the calculation. The following timing is a guide, not a script.

Minutes	Narrative	Pages here
0–7	The puzzle, axial marginal and wall condition	1–3
7–16	The cell energy and the short flux derivation	4–5
16–28	Translation, ellipse and changing lobes	6–8
28–33	Conformal computation and the three observables	9–10
33–41	Periodic theorem, proof idea and bound plot	11–12
41–45	Exact checks, limitations and discussion	13–14

For a 25-minute talk, keep the ellipse and lobe examples, state the periodic theorem at constant area, and leave the conformal coefficients and exact-state discussion for questions.

A compact dictionary.

Symbol	Meaning
$x; y = (y_1, y_2)$	Axial and transverse spatial coordinates
$\Omega(x), A(x)$	Cross section and its area
$P; p; f = p/A$	Volume density; axial marginal; section mean
$\varepsilon = D_{\parallel}/D_{\perp}$	Ratio of longitudinal to transverse diffusivity
v_n	Normal change of the boundary per unit axial distance
$\chi; u = \nabla_y \chi$	Mean-zero cell potential and its gradient
$E = A^{-1} \int_{\Omega} u ^2$	Nonnegative geometric energy at one station
$\bar{F}; \langle F \rangle_V$	Axial period average; volume average over a cell
$D^{[1]}; D_{\text{st}}[P]; D_{\text{long}}$	Local approximation; selected steady ratio; long-time coefficient

Where to read further. The accompanying `../main.pdf` contains the complete proofs: Proposition 1 and Theorem 1 for the local picture; Proposition 3 for polynomial maps; Theorem 2 and Corollary 1 for periodic bounds; Sections 7–8 and 11 for examples; Section 10 for numerical evidence. Its introduction and bibliography provide the literature comparison. The sibling `../sympy_annex/sympy_annex.pdf` records the symbolic checks and their scope.

Using these notes. Each numbered stop is designed around one idea. All eight figures are available as separate vector PDFs and PNGs for reuse in a talk. The figure-generating code, selected numerical data and build instructions accompany this document. The figures are geometric constructions or analytic plots; Figure 7 also uses existing FEM data. No particle trajectories were simulated to make the illustrations.

Open the interactive 3D explorer to orbit, zoom and pan while each animation runs. Keep the `interactive3d` folder beside this PDF. If local links are blocked, open `interactive3d/index.html` directly. The original videos remain in `animations/gallery.html`.

This companion follows the corrected 20-page manuscript dated September 25, 2026. The main paper and earlier annex are unchanged.